

20 — Solid Track: System Concept, Physics, and the Corrected Balance

v1.2 — full air conditioning from a coated-sorbent thermal swing

Function: Deliver genuine comfort (cooling **and** dehumidification) plus byproduct freshwater and hot water at DP-A, from a solid-desiccant front end coupled to a Maisotsenko-cycle (M-cycle) indirect evaporative cooler. Shared basis → doc 00; the dew-point-floor argument that makes the desiccant non-negotiable is doc 00 §3 and is not repeated here.

Status of numbers: sizing-grade unless marked; the §5 balance is the self-consistent steady state (doc 00 §4) and is **PENDING full T2** (transient model), which it supersedes in the interim.

1. Design point (reference scenario)

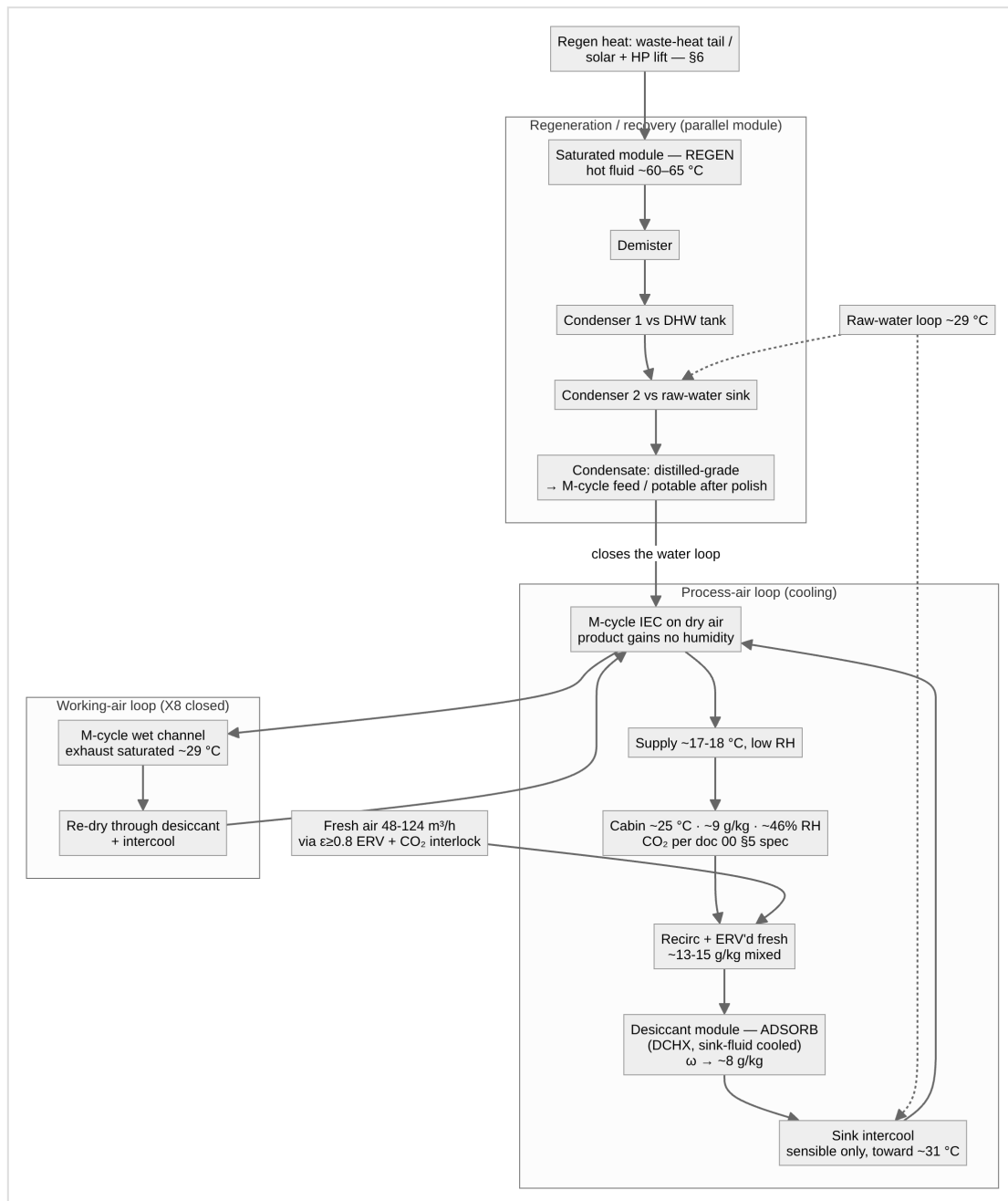
Parameter	Value	Grade
Ambient	DP-A: 32 °C / 80%, ω 24.2 g/kg, DP 28.1 °C, WB ~29.5 °C	governing (doc 00 §2)
Raw-water sink	~29 °C (tropical mixed layer nearly isothermal to 20–40 m; land sinks often cooler — margin)	sizing-grade
Conditioned volume	~100 m ³ (yacht interior / small dwelling)	platform
Occupants	4	platform
Sensible load (peak sun)	~3.5 kW	sizing-grade
Envelope + occupant latent	~1.8–2.8 kg/h (central 2.0)	sizing-grade
Total peak sorbent duty	~9–11 kg/h (self-consistent, §5)	sizing-grade, PENDING T2
Comfort target	~25 °C, ~40–55% RH	requirement

Envelope tightening remains the cheapest intervention against its term — but the envelope term is not the dominant one (§5).

2. System architecture

A desiccant + M-cycle hybrid with a heat-rejection loop doing triple duty (intake pre-cool, regeneration condensing, general rejection), plus water and heat recovery. Two or more desiccant-coated modules alternate: one adsorbing in the process path while the other is valved into the closed regeneration loop; valves swap each ~10-min half-cycle.

Air topology (adopted): recirculating process loop with **closed working-air recovery (X8, §8)** as the design intent — the M-cycle working stream is re-dried and returned rather than exhausted; ventilation is a dedicated fresh stream through its own ERV, sized for the doc 00 §5 CO₂ spec. The open cycle (working air exhausted, ambient makeup) is the commissioning/fallback mode; in open cycle the working draw comes **from the cabin** (cascade), which delivers ~600 ppm CO₂ as a structural by-product of the F1-scale makeup.



Psychrometric state chain (sizing-grade, DP-A, closed working loop):

State	T (°C)	ω (g/kg)	Note
0 Fresh (post-ERV ε 0.8)	~31	~12	pre-dried from 24.2
1 Post-desiccant	~50–55	~8	adsorption heat partly removed by sink fluid
2 Post-intercool	~30–32	~8	sensible only, to raw water
3 M-cycle product (supply)	~16.8–18	~8	ε _{dp} ~0.7 toward ~10.8 °C dew point
4 Cabin	~25	~9.1	~46% RH — steady state, gains ÷ supply flow
W Working exhaust (recycled)	~29	saturated 25.6	re-dried and returned; never vented in X8 mode

Without the desiccant the same M-cycle supplies ~31 °C air — the difference between marginal and genuine comfort is entirely the pre-dry.

3. Desorption energy — where the input goes

The dominant term is material-independent: evaporating water costs its latent heat regardless of what held it.

Energy bucket	Per litre desorbed	Driver
Latent floor	~0.67 kWh/L	heat of vaporization (fixed)
Binding excess	+0.05–0.30 kWh/L	sorbent-dependent (MOF low)

Energy bucket	Per litre desorbed	Driver
Sensible bed cycling	+0.03–0.10 kWh/L PENDING (F6)	thin coatings minimize; partly recoverable bed-to-bed — magnitude unverified, see doc 40 F6 / X14
Total	~0.8–1.0 kWh/L	latent floor is 70–85%

Of the input, only ~10–15% is recoverable **in principle**; what is recoverable *in practice* is far less, and the two figures are on different bases. Condenser latent is downhill and capped by DHW demand — ~1–3 kWh/day, which is only a few percent of the ~200 kWh/day of regeneration input the corrected \$5 duty implies (the 35+ kWh/day figure this sentence formerly compared against predates the F1 correction). Most of the heat is rejected to the sink, and **only the sensible bed-to-bed clawback actually reduces input**. Recapture order per doc 00 \$7 rule 4.

4. Ten-minute cycling — the sensible-heat caveat

Short half-cycles buy throughput from coated area (doc 22 §1) but thermally cycle the hot/cold fluid inventory and manifolds every swap; the sensible-cycling bucket above may be optimistic for the *plumbing*, not just the bed. Bed-to-bed heat-pipe recovery and manifold thermal mass minimization are design responses; M3's blank run measures the real overhead.

Scope correction (F6 / X14). The swung inventory is not only the plumbing. It is the coated-plate metal, the binder, the sorbent itself, and whatever fluid is resident in the module at swap. Per X14 the specific penalty is set by the ratio of that swung heat capacity to the water swung per half-cycle, and **half-cycle duration cancels** — shortening the cycle raises throughput per unit of coated area without changing the penalty per litre. The design levers are therefore the inert-mass-to-sorbent ratio and bed-to-bed recovery effectiveness, never cycle speed. No plate thickness, channel pitch, or per-module fluid inventory is yet specified, so the ratio cannot be evaluated: **magnitude OPEN**, gated on M3 with geometry fixed and on M1's resolution of the coated-area denominator (one-side vs two-side coating moves sorbent-per-unit-metal materially). No figure in this document is altered by this note.

5. The corrected peak balance (F1, self-consistent)

The naive sizing error: sizing to the envelope latent load (~2 kg/h) ignores where the sensible heat goes. At DP-A the raw-water sink (29 °C) cannot absorb heat from a 25 °C cabin — so **all cabin sensible load exits evaporatively via the M-cycle working air**, whose moisture must pass through the desiccant (whether exhausted-and-replaced in open cycle, or recycled in X8 mode — the drying duty is equivalent within ~10–16% because DP-A ambient is nearly as wet as saturated exhaust, 24.2 vs 25.6 g/kg).

Self-consistent steady state (doc 00 §4; Q_sens 3.5 kW, T_in 31 °C, ω_sup 8 g/kg, gains 2.0 kg/h, ε_dp 0.7):

Quantity	Value	Basis
Supply airflow	~0.43–0.51 kg/s (~1,350–1,590 m³/h)	topology-dependent (dry-draw vs cabin-draw)
Supply temperature	16.8–18.1 °C	working-air dew point
Cabin steady state	~9.1 g/kg / ~46% RH	gains ÷ supply flow — drier than target, headroom exists. (Corrected from ~40%: 9.1 g/kg at 25 °C is 46% RH. Still inside the 40–55% band, so the headroom conclusion is unchanged)
Working airflow	~0.13–0.16 kg/s (400–500 m³/h)	Q_wet ÷ Δh
Total peak sorbent duty	~9–11 kg/h	working term + fresh import + gains
Regen heat, continuous	~7.5–9 kW	~0.85 kWh/L × duty
Parasitic electrical	~0.6–1.0 kW PENDING measured ΔP	fans (supply 300–660 W + working 60–145 W at 250–550 / 150–350 Pa, η 0.32) + pumps 100–250 W

Closure — why this is coherent: the moisture pushed through the desiccant returns at the regeneration condenser as ~150–220 L/day of condensate, covering the M-cycle wet-channel feed with a ****~50–70 L/day potable-grade surplus**** (the gains + ventilation terms). Note the three quantities are distinct and only the middle one is the condensate: at 9–11 kg/h the desiccant passes **216–264 L/day desorbed**, of which ~150–220 L/day reaches the condenser train as **recovered** condensate and the balance **leaves with the ventilation exhaust**. Sizing the condenser off the desorbed figure rather than the recovered one is the safe error.

The system resolves into a **heat-driven chiller that pumps cabin heat to the sink as vapor**, with water as the internal working medium. Comfort margin note: the cabin settles drier (~46% RH) than the 55% ceiling, so a warmer supply setpoint can trade comfort headroom back into duty — a T2 optimization, with the steady-state model to run it in.

Softening conditions: milder ambient, warmer setpoint, or cabin >~31 °C (partial sink-side sensible rejection) all shrink the duty; DP-A is the continuous-duty maximum by definition.

6. The energy source decides everything

Heat source	Regen heat cost	Verdict at DP-A
Electric resistance	full price	Loses to vapor-compression AC; justified only by silence / no-refrigerant / self-sufficiency
Solar-thermal PVT direct (~45–50 °C)	free, low-grade	Equilibrium-dead at the peak point (F2): against a condensing purge (~25 g/kg) a 45–50 °C bed faces ~29–38% RH at its face — at/above AlFu's step. Zero driving force; no kinetic result rescues it. Shoulder conditions only
PVT + small heat-pump lift (to 60–65 °C)	mostly free + COP-leveraged electric	Fallback path; preserves a waste-heat-independent comfort island
Low-grade waste heat (process tail, used at 60–65 °C)	free	Primary path. Comfort + water at parasitic electrical cost

Regeneration temperature is purge-humidity-dependent — the most-misquoted figure in the sorbent literature for this duty. AlFu's "50 °C regeneration" is valid only for a *dry* purge:

Bed temp	Purge RH at bed face (vs ~25 g/kg condensing purge)	vs AlFu step (~25–30% RH)
45 °C	~38%	above step — zero driving force
50 °C	~29%	on the step — marginal
60 °C	~17%	below — works
65 °C	~13%	below — comfortable

The RH column above is computed at **~23 g/kg**, which is the condensing purge state this table was built on; the design basis is stated as ~25 g/kg and M2 is specified against a logged ~24 g/kg purge. The spread moves the 50 °C row between 29% (on the step) and 32% (above it) — i.e. exactly across the F2 boundary, which is why M2 logs the purge humidity rather than assuming it.

Design basis: **60–65 °C** open-cycle regeneration, PENDING M2 for completeness and kinetics within the 10-min half-cycle. Contrast with the liquid track's sealed still, which has no purge and no such wall (doc 00 §3, finding X2) — the structural reason the tracks layer rather than compete.

7. Water and heat-recovery strategy

Water is a byproduct of comfort, not a product to manufacture — unless heat is free. Baseline condensate is genuinely free; route it to the M-cycle wet channel, which specifically wants low-TDS feed (the M-cycle *concentrates* dissolved solids — distilled condensate is the ideal feed and protects the HMX). Deliberately making *incremental* water costs ~0.85 kWh/L versus ~4–20 Wh/L for marine RO — over-harvesting is reserved for free-heat windows. Vapor-derived condensate over an inert crystalline sorbent is intrinsically distilled-grade; polish only the potable fraction (particulate + UV; one lab test before regular drinking — doc 00 §8). Heat pipes for bed-to-bed sensible recovery and condenser→tank transport; the regenerated dry module doubles as a small **thermochemical store**; thermoelectrics rejected (~1–2% at available ΔT).

8. X8 closed working-air loop (design intent)

Route the saturated wet-channel exhaust through the desiccant and back, instead of overboard-with-ambient-makeup (doc 00 §7 rule 3):

- **Duty ≈ open cycle at DP-A** (+16% swing, partly bought back by the colder dry-draw supply and the ERV'd ventilation term) — DP-A ambient is nearly as wet as the exhaust, so closure is nearly free at the governing point. Net atmospheric enthalpy rejection of the open cycle is ≈ 0 here; size the condenser for closed-loop duty at DP-A (the maximum).
- **What closure buys:** the M-cycle becomes **water-neutral in every ambient** (the open cycle turns net water *consumer* in dry weather); sealed-envelope operation (storm/spray/dust); zero salt-aerosol ingestion on the working path (F5 mitigation); full condensate custody.
- **CO₂ correction:** closure forfeits the open cycle's incidental ~600 ppm — the dedicated fresh stream alone reads 1,920 ppm at 48 m³/h. Meet the doc 00 §5 spec by sizing ventilation (~124 m³/h ERV'd costs only +10–15% duty at this scale) and/or a TSA CO₂ bed.

9. Honest positioning

A **self-sufficiency and resilience play**, not an efficiency win over a compressor: on battery/solar electricity it costs more energy than vapor-compression AC plus RO. Its value: near-silence, no refrigerant, freshwater *byproduct* instead of freshwater *cost*, DHW, clean coupling to low-grade heat, and (X8) sealed all-weather operation. Economics improve monotonically with free heat and become compelling with an onboard/on-site waste-heat source. With F1/F2 standing, **peak-day full AC is waste-heat-coupled** (or heat-pump-assisted); the solar island — and, in the merged program, the liquid track — covers shoulder conditions and degraded operation. A conventional VC-AC remains the honest off-the-shelf benchmark and electrical backup.

10. Design principles (distilled)

1. Dehumidification is the comfort lever; the desiccant is non-negotiable (doc 00 §3).
 2. The heat-rejection loop is support infrastructure — never the drying sink.
 3. The regeneration stream is the asset: small, hot, concentrated; recover water and heat from it first.
 4. Optimize the sorbent for **desorption against humid purge**, not capture; regeneration-temperature claims are meaningless without a stated purge humidity.
 5. Model the **M-cycle working-air moisture term** explicitly — omitting it under-sizes the desiccant 3–5×; solve the steady state simultaneously (doc 00 §4).
 6. Throughput is governed by cycle time and coated area, not sorbent mass.
 7. The energy source is the master design variable.
 8. Saturated exhausts end in recovery (X8); ventilation is specified, never inherited.
 9. Redundancy outranks efficiency in the field.
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Version history

- **v1.0** — New lineage from archived 01: self-consistent §5 balance adopted (duty 9–11 kg/h, regen 7.5–9 kW, parasitic 0.6–1.0 kW); X8 closed working loop as design intent with the CO₂ correction; DP-A sole; land/marine per doc 00; ten-minute-cycling caveat recorded.
- **v1.1** — Clarifying pass from the parameter register (doc 50 §3.5), no design figure changed: cabin steady-state RH corrected from ~40% to ~46% (9.1 g/kg at 25 °C — arithmetic label only; still inside the 40–55% band and the headroom conclusion stands); §5 closure now separates desorbed / recovered / exhausted water, which were previously conflated; §3's recoverable-fraction sentence restated against the corrected F1 duty; §6 records the purge humidity its RH column was actually computed at.
- **v1.2** — §3 sensible-cycling bucket marked pending against **F6**; §4 caveat scope-corrected to the full swung inventory per X14, with the magnitude left OPEN. Waste-heat energy-source row genericised (process tail). No figure changed.